

Shubham Agrawal

Robotics Engineer | Mumbai, India

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Summary

Robotics Engineer with **3+ years** building production autonomy end to end — Linux HAL and Yocto board support up through 3D mapping, planning and control — as **top contributor** on a deployed autonomous ground robot. Now building the simulation, data and evaluation infrastructure for **Vision-Language-Action** manipulation, with model-level understanding of $\pi_{0.5}$ and **OpenVLA-OFT**.

Technical Skills

Languages: C++17, Python, MATLAB, Bash | **Practice:** GitLab CI/CD, CMake, GoogleTest, pytest, Docker, Code Coverage, Profiling
Robotics & Control: Motion Planning (D* Lite, A*, DWA), TSDF/ESDF Mapping, SLAM Integration, State Estimation, Sensor Fusion, Kalman Filtering, Robot Kinematics & Dynamics, Operational-Space & Impedance Control, Inverse Kinematics, PID, MPC, ROS, MuJoCo, Gazebo, Eigen, OpenCV

Physical AI & ML: VLA Models ($\pi_0/\pi_{0.5}$, OpenVLA-OFT), LeRobot, RLDS, Flow Matching, LoRA Fine-Tuning, JAX/Flax, PyTorch, Reinforcement Learning (PPO, SAC, TD3, TRPO), ONNX, Qualcomm QNN / Hexagon DSP

Embedded & Systems: Yocto/BitBake, Embedded Linux, HAL, GPIO, PWM, I2C, UART, ESP32, aarch64 Cross-Compilation, Shared-Memory IPC (Boost.Interprocess), Multithreading, Real-Time Scheduling, systemd

Experience

MIKO | Mumbai, India

July 2023 – Present

Robotics Engineer

Physical AI – Robotic Manipulation with VLA Models

Apr 2026 – Present

- Architected a **backend-agnostic manipulator framework** (28K+ LOC Python, 64% of commits) where one runner drives **MuJoCo** simulation or real **SO-101** arms via a config key, with scene randomization, goal predicates, a collision **safety layer** and **remote GPU inference**
- Implemented **Cartesian impedance control** (operational-space PD with task-space inertia and null-space terms) as a **C++17/Eigen** extension with **zero-copy pybind11** bindings, plus a differential **IK** solver
- Sole-authored a **LeRobot dataset curation tool** (17.6K LOC, 400 tests) cutting dataset build **22×** and preview **45×**; live analytics catch degenerate normalization and task imbalance **before** training
- Reverse-engineered $\pi_{0.5}$ and **OpenVLA-OFT** to operation level — forward pass, **flow-matching** loss, **LoRA** freeze filters — **validated against live model traces**, then **fine-tuned** $\pi_{0.5}$ on bimanual SO-101 data and wrote the team's training runbook

Autonomous Ground Robot – Navigation, Perception & Embedded Platform

July 2023 – Present

- Top contributor** to the production autonomy stack — **1,780 of 3,403 commits (52%)**, 4× the next engineer — and primary author of every core module across **28K LOC** of C++17
- Engineered an efficient algorithm to sense **free space** and **moving obstacles** from **RGB-D/depth** data by manipulating the depth image and its geometry, **patented in India**; fused bump sensors to catch glass walls
- Built **life-long 3D mapping**: **TSDF/ESDF hybrid voxel submaps** with spatial hashing, submap merging, **SLAM pose-graph integration**, **Euclidean Distance Transform** cost fields and frontier **autonomous exploration**
- Implemented **D* Lite** global planning with optimized hash and ordered-map structures, replanning over **200×** **faster than A***, and a **DWA local planner** with recovery behaviours and velocity-scaled inflation
- Created the **person-following** module (**134 of 142 commits**): a person-landmark **neural network on the Qualcomm Hexagon DSP** via **QNN**, with **NMS**, **HSV appearance re-identification** and a templated **Kalman filter** extrapolating through dropouts
- Owned the **embedded layer** on the **Qualcomm RB2 SoC**: sysfs **GPIO/PWM**, **I2C**, **UART**, encoders, **ESP32** firmware, a **PID** drive system with **IMU-fused odometry**, and a custom **Yocto** layer (45+ recipes) with Dockerized **aarch64 cross-compilation**
- Built the **ROS-free shared-memory middleware** (Boost.Interprocess, 13 typed channels) linking **6 processes**, with a **systemd** watchdog; authored the **7-stage CI/CD pipeline** gating **performance regressions** and holding **288 tests** at **93% coverage**

Education

Indian Institute of Technology (IIT) Bombay | Mumbai, India

Graduated August 2023

B.Tech + M.Tech (Dual Degree), Mechanical Engineering — CPI: 9.16 / 10.0

Projects & Achievements

- Control and Simulation of a Soft Continuum Robot** (Master's Thesis, IIT Bombay) — validated a **mass-spring model** for real-time 2D soft-robot simulation to **96%+ accuracy** and formulated an **MPC** strategy on it; fabricated a **5-segment soft continuum robot** with camera-based tracking and deployed the controller to **sub-0.1 mm** accuracy — awarded a departmental **Research Award**
- Deep RL & Robot-Learning Foundations** (self-directed) — implemented **VPG**, **TRPO**, **PPO**, **DDPG**, **TD3**, **SAC** from scratch in PyTorch with phase-gated ablations up to **BipedalWalker**, and authored a 10-module lab course on the **RLDS** robot-data standard
- Silver Medal**, Robotic Charging Challenge (Jaguar Land Rover), Inter-IIT Tech Meet 2023 — **led a team of 8** to optimize a 6-DoF arm's link geometry and generate collision-free docking trajectories | **Certificate of Appreciation**, Miko | **Technical Citation** 2023, IIT Bombay